

# Centralized MAC Protocols

TDMA

CDMA

FDMA

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OFDM  $\rightarrow$  Orthogonal Frequency Division  
multiplexing.  
Sub-carrier



OFDMA  $\rightarrow$  Orthogonal Frequency

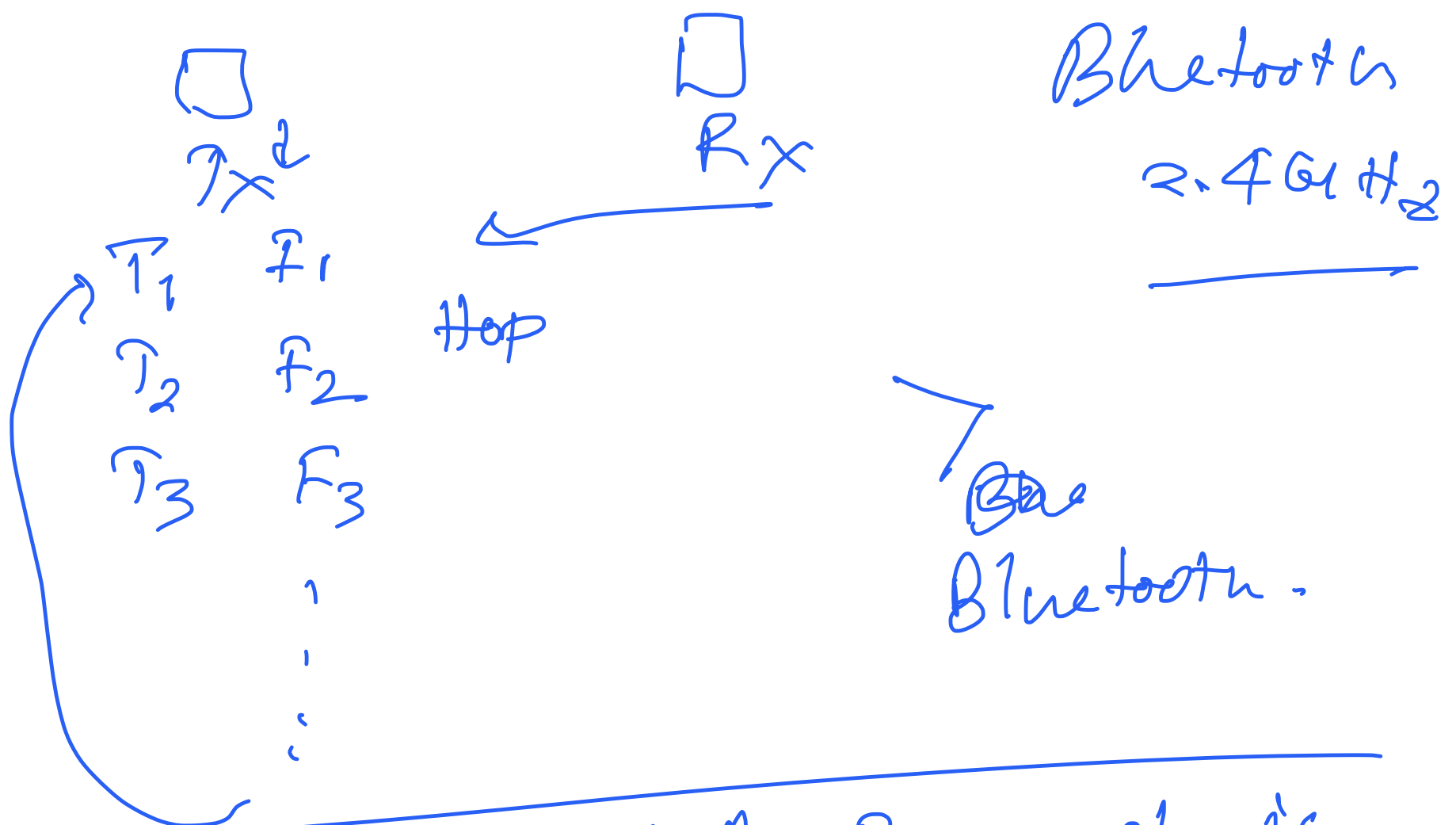
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Division Multiple Access.



How do we CSMA/CA with OFDMA?

Hopping Pattern



OFDMA  $\rightarrow$  CSMA/CA  $\rightarrow$  Each node is assigned orthogonal sub-carriers. Each node contends on the channel. abaters winning Each node transmits data using orthogonal sub-carriers so, multiple users could transmit without a collision.

OFDMA  $\rightarrow$  RVs

Each RV is assigned to each user

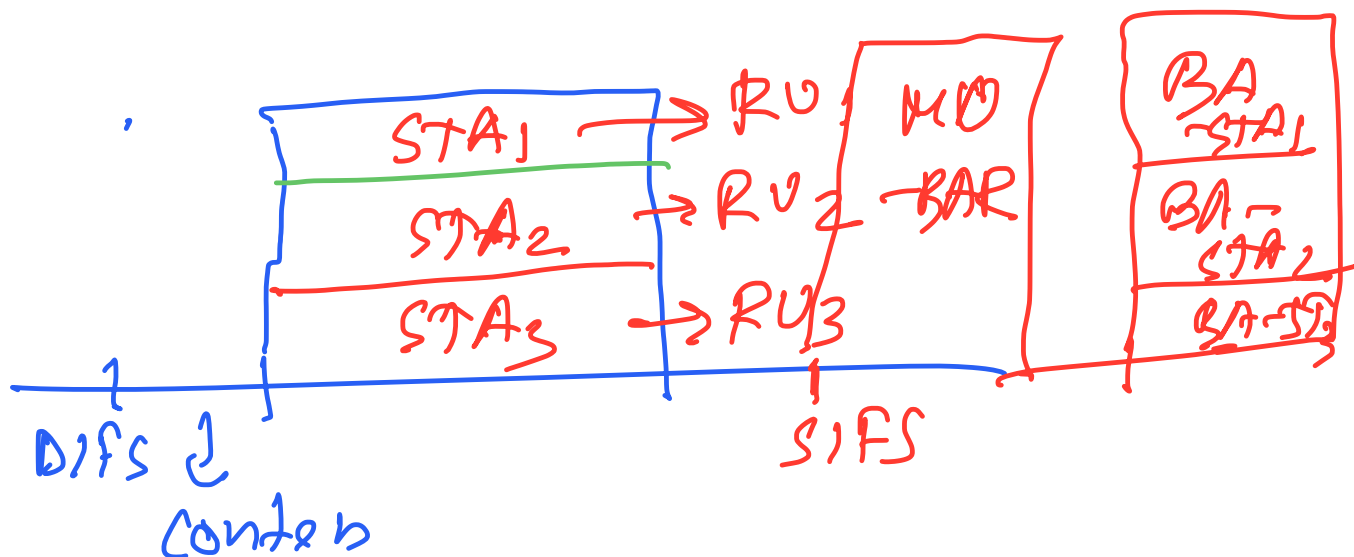
DL-OFDMA

AP transmits to multiple users at the same time

$\rightarrow$  AP first listens to the channel

$\rightarrow$  only after winning contention

AP transmits to multiple users at the same time using orthogonal sets of subcarriers or RVs (Resource Units)



Uplink OFDMA →  
Multiple nodes are transmitting to the  
AP in parallel.

